

New Hire Assignment: Create a Harbor Task

Objective

Create a Harbor task that passes both **Oracle** and **NOP** validation tests.

Resources

- Harbor Documentation: [Motivation](#)
 - GitHub Repository: [GitHub - laude-institute/harbor](#)
 - Example Tasks: Refer to `harbor_tasks/` folder in this repository
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Prerequisites

Ensure the following tools are installed before starting:

Tool	Installation
Docker	docker.com/products/docker-desktop
uv	<pre>curl -LsSf https://astral.sh/uv/install.sh sh</pre>
Cursor	cursor.sh

Windows users: Install WSL first by running `wsl --install` in PowerShell as Administrator.

Verify installation:

```
docker --version
uv --version
```

Assignment

1. Create a simple data processing task (e.g., word count, CSV filter, JSON transform)
 2. Difficulty level: Easy or Medium
 3. Requirement: Agent reads input file, processes data, and writes output file
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Task Structure

Create a folder at `harbor_tasks/<your-task-name>/` with the following structure:

```
harbor_tasks/my-task/
├── task.toml          # Metadata
├── instruction.md      # Instructions for the agent
├── environment/
│   ├── Dockerfile     # Container setup
│   └── input.txt       # Input data
├── solution/
│   └── solve.sh        # Reference solution
├── tests/
│   ├── test.sh         # Test runner
│   └── test_outputs.py # Test cases
```

Refer to existing tasks in `harbor_tasks/` for detailed examples.

Validation Commands

Install dependencies

```
uv sync
```

Oracle Test - Must return 1.0 (confirms solution works)

```
uv run harbor run --agent oracle --path harbor_tasks/<task-name> --job-name test-oracle
```

NOP Test - Must return 0.0 (confirms task does not auto-pass)

```
uv run harbor run --agent nop --path harbor_tasks/<task-name> --job-name test-nop
```

Linting

```
uvx ruff check harbor_tasks/<task-name>
```

Important Guidelines

Required	Avoid
Use absolute paths (<code>/app/file.txt</code>)	Relative paths
Compute answer in <code>solve.sh</code>	Hardcoding answers with <code>echo</code>
Include <code>memory_mb</code> and <code>storage_mb</code> in <code>task.toml</code>	Omitting these required fields
Test all behaviors from <code>instruction.md</code>	Trivial or incomplete tests

Dockerfile restrictions:

- Do not copy `tests/` or `solution/` folders
 - Do not install `pytest` in Dockerfile
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Success Criteria

- Oracle test returns 1.0
 - NOP test returns 0.0
 - Ruff linting passes
 - Pull request submitted with test output screenshots
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Submission

- `git checkout -b new-hire/<your-name>/task`
- `git add harbor_tasks/<your-task>/`
- `git commit -m "Add <task-name>: brief description"`
- `git push -u origin new-hire/<your-name>/task`

Create a pull request and include Oracle (1.0) and NOP (0.0) test outputs in the description.

Submit your assignment to deepanshu.gupta@xelron.in & developer@xelron.in in the 24 hrs of receiving this assignment.